

Supplementary Information

Permitless Concealed Carry Adoption and Firearm Mortality in U.S. States, 1999–2024

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Supplementary Note 1. Evidence hierarchy

The main manuscript interprets results through a hierarchy rather than treating every diagnostic row as an independent hypothesis test. Firearm suicide is the primary outcome. Total suicide, non-firearm suicide, firearm homicide, and total firearm deaths are secondary outcome-family checks. Firearm-law covariates, contextual covariate layers, age-adjusted outcome rates, staggered-adoption diagnostics, Arkansas recoding, leave-one-out influence, and placebo timing are sensitivity checks. Rurality, gun ownership, baseline-risk heterogeneity, and policy-feature interactions are exploratory. Cohort ATT rows, event-time rows, full legal source tables, and full robustness grids are appendix diagnostics.

Tier	Items	Interpretation
Primary outcome	Firearm suicide	Central mortality claim.
Secondary outcomes	Total suicide; non-firearm suicide; firearm homicide; total firearm deaths	Specificity and outcome-family interpretation.
Sensitivity checks	Firearm-law covariates; health-access and overdose covariates; Phase 3B2 covariate layers; age-adjusted outcome rates; staggered-adoption diagnostics; Arkansas recoding; leave-one-out; placebo timing	Tests stability and boundaries.
Exploratory checks	Rurality, gun-ownership, baseline-risk, and policy-feature heterogeneity	Hypothesis-generating only.
Appendix diagnostics	Cohort ATT rows; event-time rows; legal source table; full robustness grids	Transparency and audit trail.

Supplementary Note 2. Legal audit and policy-feature coding

The legal audit records one row per state and distinguishes clean source-verified within-panel adopters from verified non-adopters, partial cases, ambiguous cases, and baseline-permitless states. Partial, ambiguous, and baseline-permitless rows are documented so that they are not silently absorbed into the binary treatment definition.

Audit status	State count
Source-verified current adopter	26
Verified non-adopter	21
Ambiguous reviewed	1
Partial	1
Baseline permitless	1

Among the 26 clean source-verified adopter rows, 21 had a training requirement removed, 25 removed the carry-permit background-check screen, and 12 removed a permit-specific misdemeanor-violence screen. These policy-feature fields describe the carry-permit process and are treated as exploratory.

Mechanism field/value	State count
Training requirement removed: yes	21
Training requirement removed: no prior training requirement	5
Carry-permit background-check screen removed	25
Carry-permit screen removed while purchase permit retained	1
Permit-specific misdemeanor screen removed	12
Eligibility standard retained but no pre-carry check	7
Other misdemeanor-screen coding categories	7

The full state-level audit below reports the adoption year, audit status, whether the state enters the primary processed treatment map, the coding note, and source URLs used for legal verification. This table is included to make Arkansas, Mississippi, and Vermont visible rather than hiding the edge cases inside a binary treatment variable.

Table 1: State-by-state permitless concealed-carry treatment coding audit.

State	Year	Audit status	Primary map	Coding note	Source URL(s)
Alabama	2023	clean adopter	Yes	Clean source-verified within-panel adopter.	https://legiscan.com/AL/text/HB272/id/2544995 ;
Alaska	2003	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.handgunlaw.us/documents/Permitless'Ca
Arizona	2010	clean adopter	Yes	Clean source-verified within-panel adopter.	https://dps.alaska.gov/Statewide/R-I/PermitsLicensing/Firearm-FAQ ;
Arkansas	–	ambiguous reviewed	No	Excluded from primary clean-adoption map; 2021 and 2023 codings tested as sensitivities.	https://www.handgunlaw.us/documents/Permitless'Ca ;
California	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://www.azleg.gov/legtext/49leg/2r/bills/sb1108s ;
Colorado	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://www.handgunlaw.us/documents/Permitless'Ca ;
Connecticut	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://arkleg.state.ar.us/Bills/Detail?ddBienniumSess ;
					https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ;
					https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/ ;
					https://oag.ca.gov/firearms/ccw
					https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ;
					https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ;
					https://cbi.colorado.gov/firearms/concealed-handgun-permit-chp-reciprocity
					https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ;
					https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/ ;
					https://portal.ct.gov/despp/division-of-state-police/special-licensing-and-firearms/pistol-permits

State	Year	Audit status	Primary map	Coding note	Source URL(s)
Minnesota	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/ ; https://dps.mn.gov/divisions/bca/bca-divisions/administrative/Pages/Permit-to-Carry.aspx https://billstatus.ls.state.ms.us/documents/2016/html/0799/HB0786PS.htm ; https://www.handgunlaw.us/documents/Permitless'Ca https://www.senate.mo.gov/BillTracking/Bills/BillInfo https://www.handgunlaw.us/documents/Permitless'Ca https://legiscan.com/MT/bill/HB102/2021 ; https://www.handgunlaw.us/documents/Permitless'Ca https://nebraskalegislature.gov/bills/view'bill.php?Doc https://statepatrol.nebraska.gov/services/concealed-handgun-permits/chp-frequently-asked-questions
Mississippi	2015	partial	Yes	Retained in processed panel as 2015 adoption; legal audit marks the row partial.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ; https://rccd.nv.gov/FeesForms/Brady/CCW-Permit-Recognition/ https://legiscan.com/NH/votes/SB12/2017 ; https://www.handgunlaw.us/documents/Permitless'Ca
Missouri	2017	clean adopter	Yes	Clean source-verified within-panel adopter.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/ ; https://www.nj.gov/njsp/firearms/forms.shtml https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ; https://www.dps.nm.gov/concealed-carry/ https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/ ; https://gunsafety.ny.gov/pistol-permit-recertification https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ; https://ncdoj.gov/law-enforcement-training/criminal-justice/officer-certification-programs/concealed-weapon-reciprocity/ https://www.nd.gov/ndhp/node/116 ; https://legiscan.com/ND/bill/HB1169/2017 https://www.legislature.ohio.gov/download?key=1896 https://www.handgunlaw.us/documents/Permitless'Ca https://www.oklegislature.gov/cf/pdf/2019-20%20SUPPORT%20DOCUMENTS/BILLSUM/Hou https://www.handgunlaw.us/documents/Permitless'Ca https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ; https://www.oregon.gov/osp/programs/cjis/pages/oregon-concealed-handgun-license.aspx
Montana	2021	clean adopter	Yes	Clean source-verified within-panel adopter.	
Nebraska	2023	clean adopter	Yes	Clean source-verified within-panel adopter.	
Nevada	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	
New Hampshire	2017	clean adopter	Yes	Clean source-verified within-panel adopter.	
New Jersey	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	
New Mexico	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	
New York	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	
North Carolina	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	
North Dakota	2017	clean adopter	Yes	Clean source-verified within-panel adopter.	
Ohio	2022	clean adopter	Yes	Clean source-verified within-panel adopter.	
Oklahoma	2019	clean adopter	Yes	Clean source-verified within-panel adopter.	
Oregon	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	

State	Year	Audit status	Primary map	Coding note	Source URL(s)
Pennsylvania	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/state-laws/concealed-carry-in-pennsylvania/ ; https://giffords.org/lawcenter/state-laws/open-carry-in-pennsylvania/ ; https://www.pa.gov/en/agencies/psp/programs/firearms-in-pennsylvania.html
Rhode Island	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/ ; https://riag.ri.gov/i-want/get-concealed-carry-permit
South Carolina	2024	clean adopter	Yes	Clean source-verified within-panel adopter.	https://sled.sc.gov/constitutional-carry-guidance ; https://legiscan.com/SC/supplement/H3594/id/47310
South Dakota	2019	clean adopter	Yes	Clean source-verified within-panel adopter.	https://sdlegislature.gov/Session/Bill/9374 ; https://www.handgunlaw.us/documents/Permitless'Ca
Tennessee	2021	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.wvltv.com/2021/04/08/governor-lee-signs-permitless-carry-into-law/ ; https://www.handgunlaw.us/documents/Permitless'Ca
Texas	2021	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.lrl.texas.gov/legis/billsearch/billdetails.cfm ; https://sll.texas.gov/faqs/permitless-carry-guns/
Utah	2021	clean adopter	Yes	Clean source-verified within-panel adopter.	https://bci.utah.gov/concealed-firearm/ ; https://www.handgunlaw.us/documents/Permitless'Ca
Vermont	pre-panel	baseline permitless	No	Baseline permitless before the panel; not a within-panel adoption event.	https://www.nraila.org/gun-laws/state-gun-laws/vermont/ ; https://www.handgunlaw.us/states/vermont.pdf
Virginia	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ; https://vsp.virginia.gov/services/firearms/resident-concealed-handgun-permits/ ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ; https://dof.wa.gov/professional-licenses/concealed-pistol-license
Washington	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ; https://dof.wa.gov/professional-licenses/concealed-pistol-license
West Virginia	2016	clean adopter	Yes	Clean source-verified within-panel adopter.	https://wvpress.org/breaking-news/wvs-no-permit-concealed-carry-begins-today/ ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/
Wisconsin	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ; https://www.wisdoj.gov/Pages/PublicSafety/concealed-carry-weapon-license.aspx
Wyoming	2011	clean adopter	Yes	Clean source-verified within-panel adopter.	https://legiscan.com/WY/bill/SF0047/2011 ; https://www.handgunlaw.us/documents/Permitless'Ca

Supplementary Note 3. Main and secondary mortality estimates

Outcome	Coef.	SE	<i>p</i>	N
Firearm suicide	1.391	0.222	< 0.001	1222
Non-firearm suicide	0.415	0.139	0.003	1222
Total suicide	1.805	0.294	< 0.001	1222
Firearm homicide	-0.234	0.338	0.489	1142
Total firearm deaths	1.271	0.482	0.008	1222

Supplementary Note 4. Firearm-suicide diagnostic spine

The diagnostic spine places the baseline fixed-effects estimate beside population weighting, COVID-window restrictions, leave-one-adopter-out checks, placebo timing, state-specific trends, and staggered-adoption diagnostics. It is the main table for interpreting identification strength and fragility.

Table 2: **Firearm-suicide diagnostic spine.**

Diagnostic row	Estimate	Diag. p	Interpretation
Baseline unweighted fixed effects	1.391	< 0.001	Clean-adoption state-level association
Population-weighted fixed effects	0.919	< 0.001	Resident-weighted association
Pre-COVID only	1.298	< 0.001	Drops post-2019 years
COVID-excluded	1.356	< 0.001	Drops 2020–2021
External firearm-law covariates	1.096	< 0.001	Policy-environment sensitivity
Leave-one-adopter-out	1.265–1.481	–	No sign changes across adopter exclusions
Never-treated placebo timing	1.391 > 0.338	–	Observed estimate exceeds placebo timing distribution
State-specific linear trends	0.249	0.079	Central attenuation caveat
Cohort ATT windows	0.601/0.577/0.641	–	2-, 3-, and 5-year windows
Not-yet-treated dynamic ATT	0.387	–	Positive post-adoption mean
Event-time fixed effects	0.651	0.0000001	Positive post mean; diagnostic p is the minimum pre-adoption p

Estimates are deaths per 100,000 residents. TWFE rows include state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors unless otherwise noted. Modern staggered-adoption rows are reported as diagnostics because event-time checks flag pre-adoption signal; the event-time p column reports the minimum pre-adoption p rather than a post-adoption effect test.

Supplementary Note 5. Age-adjusted outcome-rate sensitivity

Crude mortality rates are the primary outcome scale because they measure realized state-year mortality burden. Age-adjusted rates are reported as a demographic-composition sensitivity check using CDC WONDER age-adjusted rates standardized to the 2000 U.S. standard population. The age-adjusted estimates preserve the crude-rate pattern: firearm suicide, total suicide, and total firearm deaths remain positive, while firearm homicide remains statistically weak and centered below zero.

Outcome	Coef.	SE	p	N
Firearm suicide, age-adjusted	1.407	0.228	< 0.001	1220
Total suicide, age-adjusted	1.914	0.327	< 0.001	1222
Total firearm deaths, age-adjusted	1.230	0.510	0.016	1222
Firearm homicide, age-adjusted	-0.230	0.426	0.590	1037

Supplementary Note 6. Firearm-law and contextual covariate sensitivity

External firearm-law covariates attenuated but did not erase the firearm-suicide estimate. Demographic-poverty covariates also preserved the firearm-suicide and total-suicide estimates. The short HRSA mental-health-provider window did not retain statistical significance and should be interpreted as a short-window sensitivity rather than a full-panel replacement. These rows are covariate-layer sensitivity checks, not uniformly superior causal models, because some time-varying covariates may lie downstream of broader state policy environments.

Firearm-suicide specification	Coef.	<i>p</i>	N
Baseline fixed effects	1.391	< 0.001	1222
External firearm-law covariates	1.096	< 0.001	1222
Health-access covariates	0.707	< 0.001	752
Overdose covariates	0.492	0.060	282
Demographic-poverty covariates	0.695	< 0.001	940
Mental-health-provider access covariates	0.116	0.744	141
Full Phase 3B2 covariates	0.176	0.633	141

Supplementary Note 7. Change-score checks

Outcome	2-year diff.	3-year diff.	5-year diff.
Firearm suicide	0.658	0.634	0.699
Non-firearm suicide	0.214	0.134	0.047
Total suicide	0.872	0.767	0.746
Firearm homicide	-0.034	0.024	0.035
Total firearm deaths	0.537	0.617	0.718

Supplementary Note 8. Staggered-adoption diagnostics

Outcome	ATT w2	ATT w3	ATT w5	Not-yet post	Interpretation
Firearm suicide	0.601	0.577	0.641	0.387	Positive post-adoption diagnostic with pretrend flag.
Non-firearm suicide	0.270	0.239	0.241	0.077	Positive post-adoption diagnostic.
Total suicide	0.871	0.816	0.882	0.464	Positive post-adoption diagnostic with pretrend flag.
Firearm homicide	-0.004	-0.107	-0.106	-0.157	No positive post-adoption diagnostic.
Total firearm deaths	0.382	0.435	0.493	0.200	Positive post-adoption diagnostic with pretrend flag.

Supplementary Note 9. Robustness and Arkansas sensitivity

Outcome	Baseline	P05 specs	Leave-one min	Leave-one max	Flag
Firearm suicide	1.391	5	1.265	1.481	Stable positive, trend-sensitive.
Non-firearm suicide	0.415	5	0.371	0.472	Stable positive, trend-sensitive.
Total suicide	1.805	6	1.656	1.954	Stable positive.
Firearm homicide	-0.234	0	-0.399	-0.049	No stable positive signal.
Total firearm deaths	1.271	4	1.058	1.532	Positive, trend-sensitive.

Arkansas recoding as 2021 or 2023 retained the sign of all five outcomes. Firearm suicide, total suicide, non-firearm suicide, and total firearm deaths retained $p < 0.05$ across both Arkansas codings; firearm homicide did not. Reintroducing Mississippi alone and in combination with both Arkansas codings preserved the positive firearm-suicide and total-suicide pattern.

Table 3: **Firearm-suicide leave-one-adopter-out estimates by excluded state.**

Excluded state	Coef.	<i>p</i>
Alabama	1.415	< 0.001
Alaska	1.405	< 0.001
Arizona	1.481	< 0.001
Florida	1.398	< 0.001
Georgia	1.413	< 0.001
Idaho	1.375	< 0.001
Indiana	1.393	< 0.001
Iowa	1.395	< 0.001
Kansas	1.414	< 0.001
Kentucky	1.451	< 0.001
Louisiana	1.411	< 0.001
Maine	1.399	< 0.001
Missouri	1.366	< 0.001
Montana	1.342	< 0.001
Nebraska	1.404	< 0.001
New Hampshire	1.391	< 0.001
North Dakota	1.358	< 0.001
Ohio	1.410	< 0.001
Oklahoma	1.383	< 0.001
South Carolina	1.406	< 0.001
South Dakota	1.403	< 0.001
Tennessee	1.416	< 0.001
Texas	1.415	< 0.001
Utah	1.424	< 0.001
West Virginia	1.420	< 0.001
Wyoming	1.265	< 0.001

Table 4: **Firearm-suicide never-treated placebo timing estimates.**

Placebo shift	Coef.	<i>p</i>
0	-0.026	0.850
1	-0.246	0.173
2	-0.062	0.788
3	0.081	0.721
4	0.385	0.060
5	0.291	0.047
6	0.054	0.675
7	0.050	0.767
8	-0.032	0.883
9	-0.224	0.375
10	-0.275	0.156

The absolute median placebo estimate was 0.081 deaths per 100,000 and the absolute 95th percentile was 0.338. The observed firearm-suicide estimate was 1.391, exceeding that placebo timing distribution.

Supplementary Note 10. Heterogeneity and policy-feature checks

Rurality was the clearest recurring heterogeneity dimension: firearm suicide, non-firearm suicide, and total suicide had positive high-rurality interactions. These models are exploratory because state-level subgroup indicators are coarse and adoption is politically patterned.

Outcome/dimension	Interaction	<i>p</i>
Firearm suicide, high rurality	0.988	< 0.001
Total suicide, high rurality	1.733	< 0.001
Non-firearm suicide, high rurality	0.745	< 0.001
Non-firearm suicide, high gun ownership	0.586	< 0.001

Policy-feature heterogeneity did not identify a stable mechanism. None of the 15 training, carry-permit background-check, or misdemeanor-screen interaction rows had $p < 0.05$. Five rows were sparse or unavailable because the feature-specific clean-adopter comparison group was absent.

Supplementary Note 11. Homicide missingness audit

Firearm-homicide outcome rows have fewer observations than suicide and total-firearm rows because some state-year rates are unavailable or suppressed. The missingness audit shows that all missing firearm-homicide rows occur in the smallest-population state-years.

Table 5: **Firearm-homicide missingness audit.**

Group	Possible	Observed	Missing	Missing pct.
All state-years	1222	1142	80	6.5%
Adopter-state state-years	676	610	66	9.8%
Non-adopter-state state-years	546	532	14	2.6%
Q1 smallest	312	232	80	25.6%
Q2	312	312	0	0.0%
Q3	286	286	0	0.0%
Q4 largest	312	312	0	0.0%

Missing firearm-homicide rates are concentrated in the smallest-population state-years. No state-year in population quartiles 2–4 is missing the firearm-homicide rate.

State	Missing	Possible	Missing pct.
New Hampshire	16	26	61.5%
North Dakota	15	26	57.7%
South Dakota	14	26	53.8%
Wyoming	14	26	53.8%
Hawaii	13	26	50.0%
Maine	6	26	23.1%
Delaware	1	26	3.8%
Montana	1	26	3.8%

Supplementary Note 12. Full diagnostic source files

The following repository outputs provide the full machine-readable diagnostic trail used to write the manuscript and this supplement:

- `outputs/tables/did/twfe.did.main.results.csv`
- `outputs/tables/did/twfe.did.ageadj.sensitivity.summary.csv`
- `outputs/tables/did/twfe.did.firearm.law.control.summary.csv`
- `outputs/tables/did/twfe.did.nonfirearm.confounder.summary.csv`
- `outputs/tables/did/twfe.did.phase3b2.confounder.summary.csv`
- `outputs/tables/main/welch.change.score.results.csv`

- `outputs/tables/modern_did/modern_did_summary.csv`
- `outputs/tables/modern_did/event_time_never_control_estimates.csv`
- `outputs/tables/robustness/robustness_summary.csv`
- `outputs/tables/robustness/leave_one_state_out.csv`
- `outputs/tables/robustness/placebo_never_treated.csv`
- `outputs/tables/robustness/arkansas_treatment_sensitivity_summary.csv`
- `outputs/tables/heterogeneity/heterogeneity_did_results.csv`
- `outputs/tables/mechanism/mechanism_heterogeneity_results.csv`
- `data/policy/permitless_carry_legal_audit.csv`